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Self-organized criticality in the Abelian sandpile

Driving, dissipation, and avalanche statistics

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Abstract

Adding one grain to a sandpile can leave it stable or trigger an avalanche across many sites. This project uses the two-dimensional Bak–Tang–Wiesenfeld model to investigate how that variation arises from a fixed local rule. We compare uniform input with spatially biased input, cylindrical boundaries and four-grain batches. We simulated 10,000,000 input grains across four baseline sizes and five seeds per condition, using independently checked implementations. At $L = 64$, the uniform open model had mean height 2.10361 and mean topplings 153.033 per grain, compared with the finite-grid prediction 153.043. Bias and cylindrical boundaries increased mean topplings to 266.669 and 357.511, while mean loss remained close to one per grain. A small height difference under biased input was not reproduced in separate, longer runs. All 500,000 matched batches reproduced grouped sequential topplings and loss exactly, although their event distributions changed. Normalized discrete fits favour cutoff descriptions over pure powers on the chosen finite ranges. These comparisons distinguish changes in the stationary state measure, the response to input and the grouping of avalanche events.

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Chapter 1

Introduction

Each input to the sandpile is just one grain, but its effect depends on the configuration it lands on. A stable site may absorb the grain, or a toppling may push neighbouring sites past their thresholds and start a cascade. This makes the model useful for studying a basic question in complex systems: how can the same small input produce responses of very different sizes?

Bak, Tang, and Wiesenfeld introduced self-organized criticality (SOC) to describe systems that develop critical behaviour under slow driving, without an externally tuned critical parameter [1, 2]. Their sandpile model combines gradual input with rapid redistribution and loss at the boundary. Its simple rules allow us to examine how local instability affects the system as a whole.

We ask whether changes in avalanche statistics reflect a different stationary state, a different response to external driving, or simply a different grouping of avalanche events. Using the two-dimensional BTW model in the assignment [7], we examine relaxation, height, flux, distributions and correlations across lattice sizes. The Abelian property provides exact checks on the simulation.

Three controlled extensions separate these possible sources of change. Spatial bias changes where grains enter while preserving the stationary recurrent measure on the same graph. Cylindrical boundaries change connectivity and dissipation, potentially changing both stationary configurations and responses. Four-grain batches regroup a matched input stream; exact Abelian identities preserve sequential toppling totals, loss totals and final configurations. Each comparison changes one feature while retaining deterministic threshold-four toppling, so a change in event statistics can be interpreted in terms of its specific mechanism.

Chapter 2

Model, observables, and physical interpretation

2.1 Dynamics and observables

Let $\Lambda_L = \{0, \dots, L-1\}^2$ be the lattice, with stable heights $h(x) \in \{0, 1, 2, 3\}$. Starting from an empty grid, the primary baseline chooses a site independently and uniformly, adds one grain and lets the grid relax completely. A legal elementary toppling requires $h(x) \geq 4$, subtracts four grains and transfers one in each axial direction. Grains sent outside the lattice leave the system. The threshold remains four at edges and corners. The next grain is added only after stabilization, and external time counts input events.

Let $u_t(x)$ count elementary topplings at site x in event t. The sitewise toppling-count vector u_t is called the odometer. Let X_t be the set of input sites. We measure

$$S_t = \sum_x u_t(x), \qquad A_t = \sum_x \mathbf{1}_{\{u_t(x)>0\}}, \tag{2.1}$$

$$D_t = \sum_x [4 - d_G(x)] u_t(x), \qquad \ell_{X_t} = \begin{cases} 1 + \max\limits_{u_t(x)>0} \min\limits_{s \in X_t} d_G^{\mathrm{geo}}(x, s), & S_t > 0, \\ 0, & S_t = 0. \end{cases} \tag{2.2}$$

Here $d_G(x)$ is the number of internal directional neighbours and d_G^{geo} is graph distance. We use B for adjacency, distinct from avalanche area A. On the open square, distance is Manhattan distance. With one input site, reach is the greatest distance from that site to a toppled site, plus one. This gives a single-site avalanche reach one and measures spatial extent. For several input sites, each toppled site is measured from its nearest source, so the interpretation depends on the source set.

All events, including $S = 0$, enter per-input means and mass balance. The main avalanche distributions and correlations condition on $S > 0$ and retain $D = 0$; positive-loss fits condition on $D > 0$. We report the zero-event probabilities separately. Height frequencies are sampled every 128 added grains after stabilization, including after zero events. For batches, elapsed input is measured in grains as well as events. The distance from a batch to a dissipative boundary is the minimum over its input sites.

2.2 Scale invariance and the SOC mechanism

Scale invariance means that changing scale leaves the shape of a relation unchanged, apart from an overall factor. Suppose a positive function satisfies $f(cx) = g(c)f(x)$ for every $c, x > 0$. Rescaling first by a and then by b gives $g(ab) = g(a)g(b)$. Under continuity or suitable measurability, $g(c) = c^{-\alpha}$, giving

$$f(x) = f(x_0)(x/x_0)^{-\alpha} = Cx^{-\alpha}. \tag{2.3}$$

Invariance under a single discrete factor also allows forms such as log-periodic functions. A probability distribution needs normalizable support as well. For an unbounded discrete power tail with $\alpha > 1$, $\mathbb{P}(X \geq n) \sim C' n^{-(\alpha-1)}$, so the PMF and CCDF have different exponents. On a finite lattice, microscopic and system-size cutoffs limit the range over which scaling can hold approximately.

The separation between slow input and rapid relaxation is central to the BTW model. Local thresholds allow activity to spread, interior topplings conserve grains and the boundary removes them.

These features motivate examining height, transport and avalanche distributions together. A power-law description requires a stated fitting range and distributional evidence beyond a straight segment on logarithmic axes [3].

2.3 Boundaries, steady states, and infinite volume

If m_t grains enter an event, total mass satisfies

$$M_t = M_{t-1} + m_t - D_t, \qquad \overline{D} = \overline{m} - \frac{M_T - M_0}{T}. \tag{2.4}$$

Open boundaries allow grains to leave as new grains arrive. Every site connects to a sink, and stable mass is bounded by $3L^2$. The endpoint term in Eq. (2.4) therefore vanishes over a long window, even without a mixing argument. Average loss can approach average input while other statistics still show transients. Without dissipation, continued input to a finite system eventually exceeds the capacity of any stable configuration.

Starting from an empty infinite lattice, any finite collection of grains can stabilize by spreading outward [9]. Continued input in a finite source region increases the total mass and expands the affected region. Individual finite additions can still stabilize. An occupied infinite background or an infinite-volume stationary measure requires additional assumptions.

We use *macroscopic steady state* to mean that selected bulk observables show no systematic drift. *Statistical stationarity* is stronger: the joint probability laws remain unchanged under shifts in time. In the physical sense used here, *equilibrium* has no sustained externally driven transport; detailed balance is a common criterion in stochastic models. The relevant description of the driven sandpile is a nonequilibrium stationary state. Grains continue to enter and leave even when the height statistics settle.

Chapter 3

Abelian structure and exact benchmarks

3.1 Matrix conditions and the threshold

Using the handout's negative-diagonal convention,

$$\Delta = -4I + B, \qquad K = -\Delta = 4I - B, \qquad h \longmapsto h - Ke_x. \tag{3.1}$$

Nonnegative off-diagonal entries of Δ ensure that a toppling adds grains to other sites or leaves them unchanged. Symmetry describes the undirected transfers used here. The negative diagonal removes grains from the toppled site, while nonpositive row sums prevent mass creation. A strictly negative total row sum means that some mass can dissipate. For the open square, row sums are zero in the interior, -1 at non-corner edges and -2 at corners for $L \geq 2$, giving a total of $-4L$. The single-site case has row sum -4. The threshold is $k = -\Delta_{xx} = 4$.

Sink connectivity supplies the further condition needed for stabilization: every site must have a path to a sink. Dissipation elsewhere cannot stabilize a disconnected component with no such path. Both graphs used here meet this condition. Symmetry describes these graphs but is not required for every generalized Abelian sandpile. The cylinder wraps horizontally and has sinks only at its top and bottom; its total row sum is $-2L$ at the lattice sizes studied.

Two inconsistencies in the handout affect the local rule and its example. Equation (6) requires $h(x) \geq -\Delta_{xx}$ in place of $h(x) \geq \Delta_{xx}$. In Figure 3, the upper-middle toppling omits a grain transferred to the centre. Applying the stated local rule gives

$$\begin{pmatrix} 1 & 3 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 2 & 2 \end{pmatrix}, \quad (S, A, D, \ell) = (4, 4, 4, 3). \tag{3.2}$$

The input mass is 19 and the final mass is 15. The initial array already includes the added grain.

3.2 Termination and unique stabilization

The Abelian property means that choosing a different legal toppling order leaves the stabilized configuration unchanged. We first establish termination, then show that every complete legal sequence has the same toppling counts. The argument follows the finite-graph treatment [6, Theorem 2.3 and Exercise 2.6] and Dhar's Abelian structure [5].

Termination. Attach missing neighbours to an absorbing sink. For a real vector z,

$$z^\top K z = \sum_{\{x,y\} \text{ internal edge}} (z_x - z_y)^2 + \sum_x [4 - d_G(x)] z_x^2. \tag{3.3}$$

Sink connectivity makes this expression positive for nonzero z, so K is invertible. Let $Kw = \mathbf{1}$. If w had a nonpositive minimum at x, then $(Kw)_x \leq [4 - d_G(x)]w_x \leq 0$, contradicting $(Kw)_x = 1$. Hence $w > 0$. The potential $\Phi(h) = w^\top h$ is nonnegative for nonnegative heights, and each elementary toppling changes it by

$$\Phi(h - Ke_x) - \Phi(h) = -w^\top K e_x = -1. \tag{3.4}$$

The potential drops by one at each toppling and stays nonnegative. Only finitely many topplings are therefore possible, and continuing while an unstable site exists must reach a stable configuration. This applies to both the open graph and the dissipative cylinder.

Local commutation. Suppose distinct sites x, y are both unstable. Toppling either site leaves the other unstable, so both orders are legal and

$$T_y T_x h = h - Ke_x - Ke_y = T_x T_y h. \tag{3.5}$$

The simultaneous-instability condition ensures legality in both orders. Conditional operators that do nothing at stable sites need not commute on arbitrary configurations.

Unique stabilization. Let u, v count topplings in two complete legal stabilizations of the same configuration h. Follow the first sequence to the proposed toppling that would first exceed v at a site x. Just before it, the partial counts satisfy $a_x = v_x$ and $a_y \le v_y$ at every other site. Consequently,

$$(h - Ka)_x = h_x - 4v_x + \sum_y B_{xy} a_y \le h_x - 4v_x + \sum_y B_{xy} v_y = (h - Kv)_x \le 3. \tag{3.6}$$

Site x would already be stable, contradicting the legality of the proposed toppling. Hence $u \le v$. Reversing the argument gives $u = v$, so both sequences have the same final configuration. Every observable determined by the odometer is therefore independent of the schedule.

3.3 Stationary measures, ergodicity, and mean response

On a finite graph connected to a sink, positive addition probability at every site makes the dynamics eventually enter a single recurrent class with probability one. Its stable configurations can reach one another through additions, and subsequent additions cannot leave the class. Stable configurations outside it are transient. Each addition permutes the recurrent class, and any probability mixture of the permutations preserves its uniform measure [6, Theorems 2.9, 2.11; Exercise 2.16]. Changing the full-support driving distribution on the *same graph* therefore preserves stationary height statistics. Uniformity applies to recurrent configurations, not all stable configurations; individual-site heights remain correlated and need not be equiprobable.

Time averages are ergodic on this finite recurrent class. Convergence of the distribution at a fixed time also requires aperiodicity. The one-site model, for example, cycles through four heights; sampling every four additions preserves its initial phase. This is why we compare time and ensemble estimates at specified sampling ages and use the recurrent-state theory for the ergodic statement.

Before simulation, the naive prediction assigned equal probabilities to heights zero through three, giving mean 3/2. The literature value 17/8 is the infinite-volume recurrent mean [8]. Finite grids have their own response benchmark. For the complete input vector a_t, balance gives

$$h_t = h_{t-1} + a_t - Ku_t, \tag{3.7}$$

$$K\mathbb{E}[u_t] = \mathbb{E}[a_t] = mq, \qquad \mathbb{E}[S_t] = \mathbf{1}^\top K^{-1}(mq), \qquad \mathbb{E}[D_t] = m. \tag{3.8}$$

Under stationarity, this is the weighted form of Dhar's mean-toppling formula [6, Theorem 2.18]. We solve sparse linear systems and check their residuals, avoiding an explicit inverse. An exact identity for a finite observation window is

$$\sum_t S_t = \sum_t w^\top a_t + w^\top (h_0 - h_T), \qquad Kw = \mathbf{1}. \tag{3.9}$$

The endpoint term accounts for mass still stored in the grid, so this identity can check a realized input stream without assuming stationarity.

Chapter 4

Computation and experimental design

4.1 Independent implementations and validation

A reference simulation scans the grid and performs one elementary toppling at a time. The faster algorithm keeps a queue of unstable sites, performs $\lfloor h(x)/4 \rfloor$ consecutive legal topplings at each visit and adds that count to the odometer. Each site has at most one pending place in the queue, avoiding repeated scans of stable sites. Both Python and C versions were compared with the reference.

Mass changes were checked against the grains added and lost after every event. Height frequencies provided an independent check of the actual grid mass every 128 added grains, supplemented by direct sums of all site heights during each run. A separate array-based simulation checked the actual mass after every event in validation runs. Heights and counts use signed 64-bit integers, and repeated batch locations receive every grain assigned to them.

The checks cover all $256 \times 4 = 1,024$ stable 2×2 additions and 1,000 bounded random configurations for each of the Python and C implementations. Forward, reverse and random legal toppling orders give identical results. Further checks include rectangular and degenerate grids, repeated topplings, all four observables and the corrected handout example. Periodic connections, dissipation and distances on the cylinder were checked, as were biased-input probabilities and frequencies, and batch odometer sums and area unions. Exact enumeration gives 192 recurrent 2×2 states and verifies the uniform invariant measure for uniform and unequal positive input probabilities, including four successive additions.

Known discrete distributions, sparse samples and autoregressive series were used to check the statistical calculations. Fits compare several initial parameter choices and check the best converged solution against an independent projected-gradient calculation. All 144 aggregate fits passed these checks.

4.2 Profiling and performance

Profiling identified repeated neighbour construction as a major cost, which was removed by storing the neighbour lists. A later $L = 64$ profile placed most time in the active-site queue. Implementing this algorithm in C, with Python generating the input sites, reduced that cost. Observables are calculated as sites topple. Reach calculations depend on the number of toppled sites and input locations; each batch here has four locations.

Implementation	Median (ms)	Range (ms)	Speed ratio
Elementary scan, Python	552.216	[531.345, 562.245]	1.0
Bulk queue, Python	107.030	[106.292, 107.353]	5.2
C queue, one event per call	95.477	[94.000, 107.416]	5.8
C queue, full input sequence	0.167	[0.166, 0.205]	3298.7

Table 4.1: Stabilization times for the same $L = 16$ calculation: 2000 predetermined inputs from an all-three grid, giving 25377 elementary topplings. Each implementation first completes one untimed run. Each of five timing samples then repeats complete calculations for at least 0.2 s, resetting the grid each time; reported times are per calculation. All methods give identical S, A, D, ℓ and final grids. Compilation (0.477 s) and initial loading/warm-up (0.751 s) are timed separately. Ranges are minimum to maximum; speed ratios use the Python scan median. Times exclude generating input sites and reading or writing data.

The Python queue is 5.2 times faster than the scan (Table 4.1), combining active-site selection,

consecutive topplings and stored neighbour lists. Processing the full input sequence in C is faster still than calling the C routine after each event, because it reduces calls between languages. Each timing sample lasts at least 0.2 seconds, although the C time per complete calculation is below one millisecond. The environment used Python 3.12.2, NumPy 1.26.4, SciPy 1.13.0, Matplotlib 3.8.4 and Apple Clang 17 (arm64 macOS, eight logical CPUs).

4.3 Simulation lengths and initial conditions

The preliminary runs used $L = 64$ for open uniform, open biased and cylindrical input, each starting from empty, random stable and all-three grids with independent random-number sequences. They followed $60L^2$ grains in the uniform and cylindrical cases and $100L^2$ under bias. A separate equilibrated, matched batch comparison checked 81,920 subsequent grains. Based on these runs, we used five seeds per condition, baseline sizes $L = 8, 16, 32, 64$, extension sizes $L = 32, 64$, and 200,000 measured grains per seed. Burn-in is $20L^2$ for baseline and cylinder, and $40L^2$ for bias. Batches start from the matched baseline's grid after burn-in.

The 50 main runs contain 10,000,000 measured grains and 8,500,000 events, plus 2,080,000 burn-in grains. Each batch run has 50,000 events and uses the same initial grid and subsequent input sites as its single-grain control. Those two million batch grains form paired comparisons. Preliminary runs, main runs, toppling-order checks, the 100 independent $L = 32$ ensemble chains and block resampling use independent random-number sequences. The main simulations took 40.31 s, including 24.25 s in simulation calculations and 14.28 s in data storage. These times include burn-in.

4.4 Uncertainty and distribution fitting

We estimate primary 95% Student-t intervals from five independent seed estimates, with four degrees of freedom. Each seed is one replicate, which accounts for dependence within a run but gives limited precision with five replicates. Independent extension differences use Welch intervals; batch comparisons use exact pairing. The time-series comparison uses 24 complete contiguous blocks of 8,192 grains, covering 196,608 grains. The remaining grains are retained in all-event summaries. Block lengths of 4,096 and 16,384 grains and 200 block-bootstrap resamples assess sensitivity. Autocorrelation is calculated separately for height, topplings, loss, dissipation indicators and height frequencies. The initial-positive-pair time estimate uses a finite lag range and has a floor of one, which short-lag anticorrelation can reach.

Fits use the exact counts of raw integer observations, with normalized likelihood

$$p(x \mid \alpha, \lambda) = \frac{x^{-\alpha} e^{-\lambda x}}{\sum_{n=x_{\min}}^{x_{\max}} n^{-\alpha} e^{-\lambda n}}. \tag{4.1}$$

The candidates are power ($\lambda = 0$), cutoff power and exponential ($\alpha = 0$), fitted to identical observations and support. Numerical bounds are $-10 \le \alpha \le 10$ and $0 \le \lambda x_{\max} \le 10^5$; no fitted free parameter reaches them. Finite support allows negative effective slopes. Each seed must retain at least 1,000 observations. The fixed lower limits are $(16, 8, 1, 3)$ for (S, A, D, ℓ); sensitivity checks halve and double them, with a minimum of one. For S, the upper fitting limit L^2 selects an intermediate range. The other upper limits, L^2, $3L^2+1$ and $2L-1$, are physical bounds. We record observed maxima separately from these fitting limits.

Parameter intervals are calculated across the five seed fits. KS distances describe discrepancies between fitted and empirical CDFs. Marginal log-score differences and their block sensitivity use the fitting data, so these comparisons are descriptive; the cutoff model also has an extra parameter. For logarithmic displays, counts are divided by bin width and sample count. Fits use the original integer observations, with each size and model analysed separately.

Chapter 5

Results

5.1 Stationarity, height, and sampling

5.1.1 Approach to the measured regime

The pilot resolves initial mass accumulation and later fluctuations (Figure 5.1). Empty starts give a lower mean in the first $10L^2$-grain window. The longest estimated height or frequency correlation time was about 1,195 grains, making the chosen block about seven times longer. These diagnostics guided burn-in and blocking for the measured observables. Each starting state had one independent pilot chain.

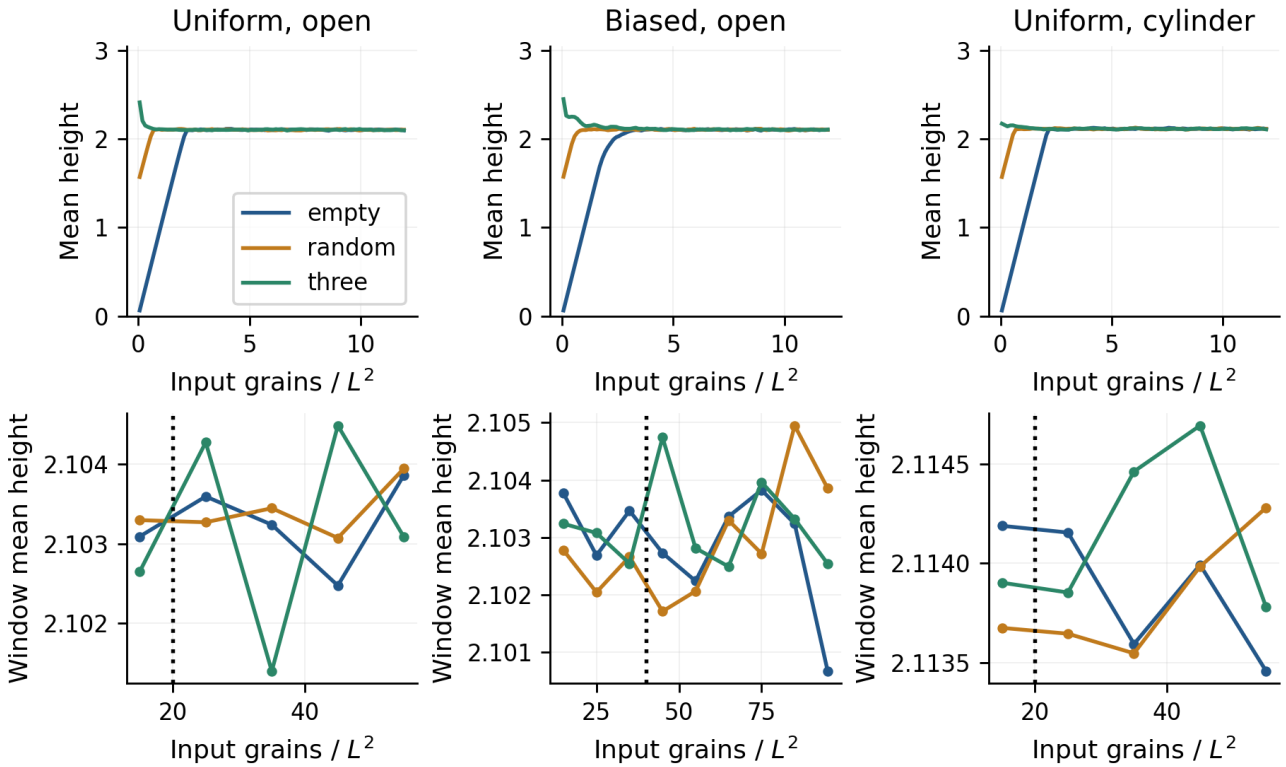


Figure 5.1: Different initial configurations approach similar late-stage height levels within each model, supporting the selected burn-in for the measured observables. Independent $L = 64$ pilots start empty, random stable and all-three. Top: 512-grain mean heights over the first $12L^2$ grains. Bottom: connected $10L^2$-window means after the first window; vertical lines mark selected burn-in. Uniform/cylinder pilots last $60L^2$ grains; biased pilots last $100L^2$. The main runs use independent seeds. These checks do not establish complete mixing.

Across four consecutive measurement windows, height, all four height frequencies, mean topplings, loss and dissipation probability fluctuate without a common continuing drift. The small height difference under bias prompted a separate investigation (Section 5.4.1). Every main run passed the exact mass and potential-balance checks in Eq. (3.9), and every sparse benchmark residual was below 10^{-10}.

5.1.2 Height, finite size, and flux

The equal-height-frequency prediction of 3/2 is inconsistent with the measured baseline (Table 5.1, Figure 5.2). Mean height rises from 1.98211 at $L = 8$ to 2.10361 at $L = 64$, approaching the infinite-

volume value 17/8. Height-three sites are much more frequent than height-zero sites.

L	Mean height	Pred. S	Measured S	$P(S=0)$	$P(D=0 \mid S>0)$	$P(D>0)$
8	1.98211 ± 0.00114	3.462	3.463 ± 0.003	0.6194	0.2331	0.2919
16	2.04645 ± 0.00024	11.338	11.333 ± 0.018	0.5910	0.3883	0.2502
32	2.08355 ± 0.00044	40.580	40.611 ± 0.066	0.5743	0.5245	0.2024
64	2.10361 ± 0.00025	153.043	153.033 ± 0.426	0.5649	0.6373	0.1578

Table 5.1: Baseline simulations with five seeds and one million grains per size. Heights and topplings are means ± 95% seed-level interval half-widths, averaging over all input events. Predictions come from the sparse systems. The last columns give zero-avalanche probability, zero-loss probability among avalanches, and loss probability per input.

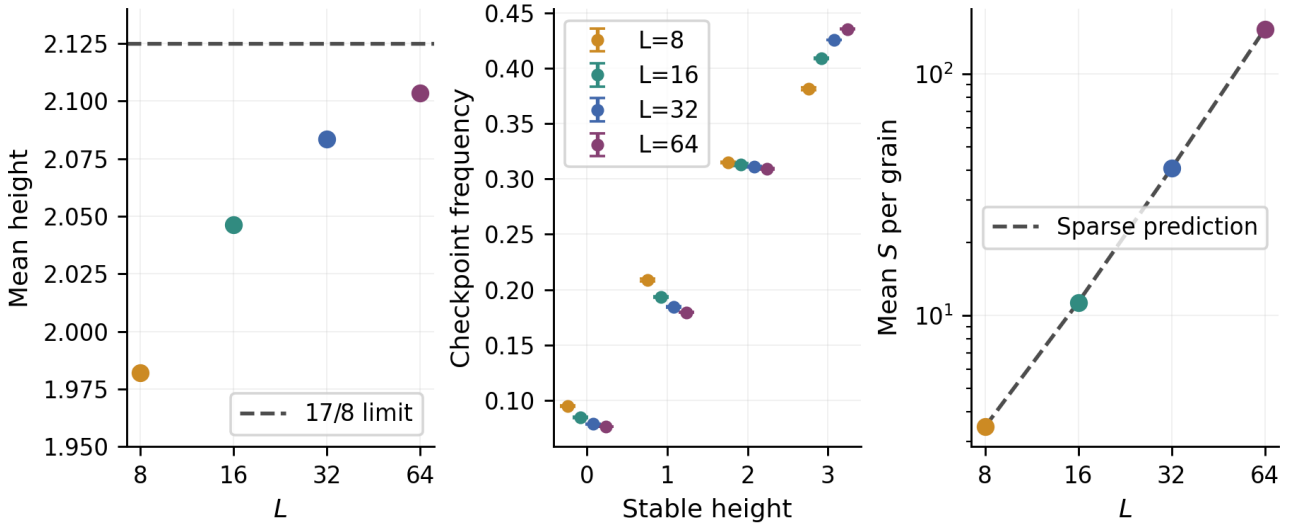


Figure 5.2: Stable heights have unequal frequencies, mean height increases with L, and measured mean topplings agree with finite-grid predictions. Each size uses one million measured inputs across five seeds. Left: mean height and the 17/8 infinite-volume reference. Centre: height frequencies after stabilization at regular intervals. Right: all-input mean topplings and independent predictions. Error bars are 95% seed intervals; some are smaller than the markers.

At $L = 64$, mean topplings are 153.033, with interval $[152.606, 153.459]$ containing the prediction 153.043. All four baseline sizes agree with their finite-grid benchmarks at the reported precision. Mean loss per grain is 0.9998, with interval $[0.9996, 1.0000]$; the endpoint mass accounts exactly for its small departure from one. As L increases, loss occurs less often but dissipative events lose more grains: $\mathbb{E}[D \mid D > 0]$ rises from 3.426 to 6.337. Larger grids carry roughly the same mean flux through rarer, larger loss events.

5.1.3 Time versus independent ensemble

For $L = 32$, one long trajectory gives time-sampled mean height 2.08420, interval $[2.08358, 2.08482]$, and height-three fraction 0.42566, interval $[0.42520, 0.42613]$. Sampling 100 independent chains after $20L^2$ inputs gives 2.08451, interval $[2.08182, 2.08720]$, and 0.42676, interval $[0.42445, 0.42906]$, respectively. The intervals overlap. At the earlier ages of L^2 and $2L^2$ inputs, the ensemble summaries still show the effect of starting empty.

The time and ensemble estimates remain consistent when the block length changes. Height has persistent positive dependence, whereas topplings show short-lag anticorrelation. These different patterns support using separate dependence diagnostics for each observable.

5.2 Avalanche distributions and finite-range fits

At $L = 64$, the largest observed toppling count is 19687, above the fitting cap $L^2 = 4096$. These larger events remain in the means and empirical distributions (Figure 5.4). The bounds L^2 for area and $2L - 1$ for single-source open reach reflect the finite grid directly. Loss measures transport to the boundary and differs in distribution from topples or area.

Zero events make a substantial part of the sample. At $L = 64$, $P(S = 0) = 0.5649$, and $P(D = 0 \mid S > 0) = 0.6373$. A positive-loss tail describes only the dissipative subset of avalanches.

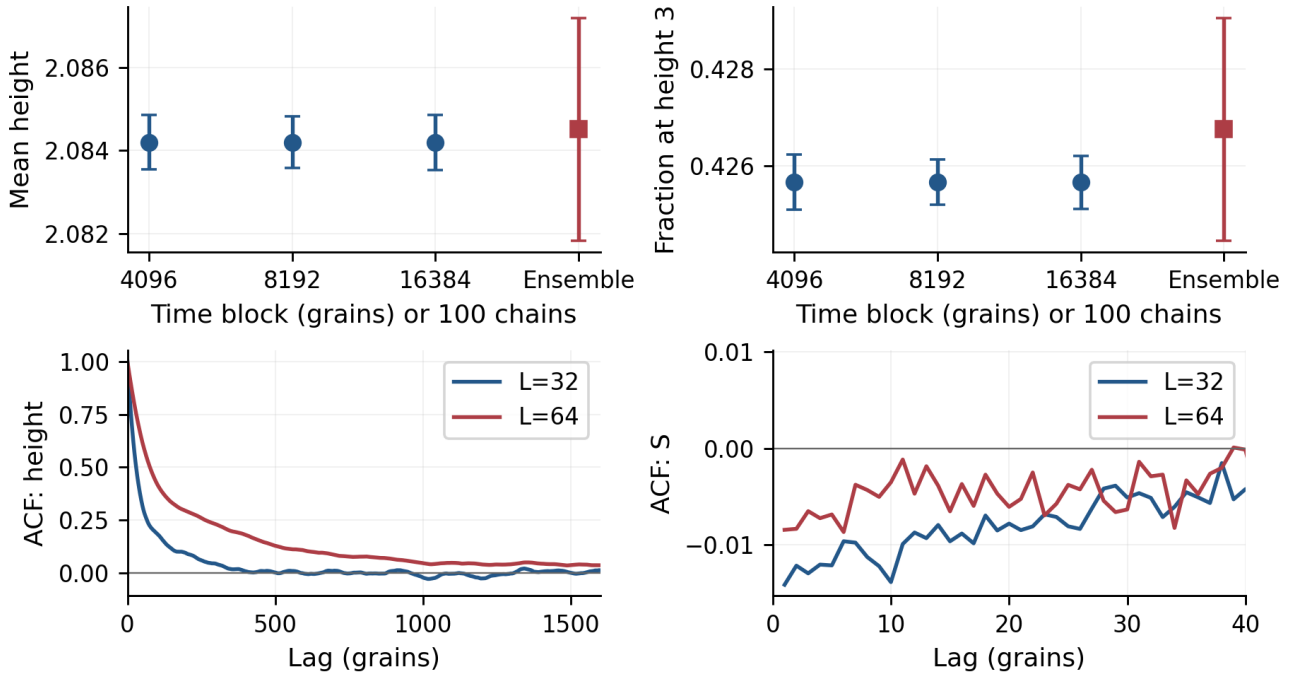


Figure 5.3: Time and ensemble sampling. Top: estimates and 95% intervals from one $L = 32$ trajectory at three block lengths, compared with 100 independent chains at age $20L^2$. Time estimates cover 196,608 grains, with height-three frequency sampled every 128 grains. Bottom: height and toppling ACFs for the first run at $L = 32, 64$. The different dependence patterns motivate separate diagnostics for each observable. The ensemble unit is an independent chain.

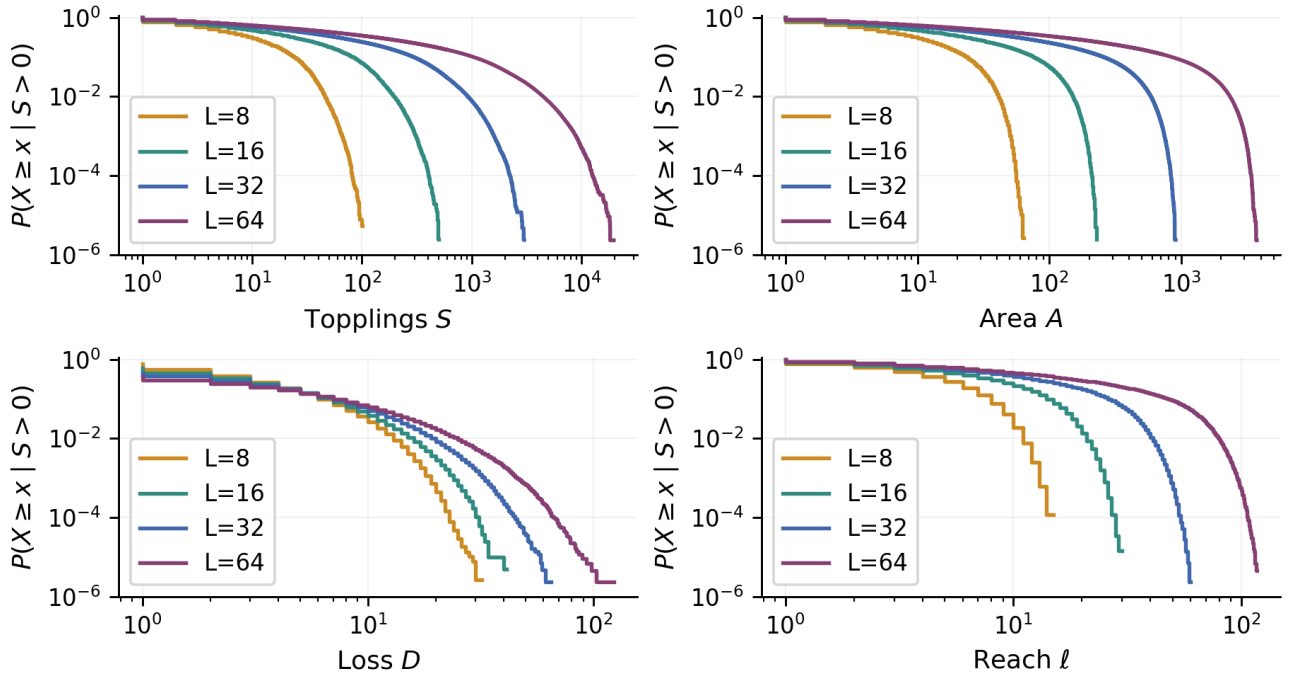


Figure 5.4: Avalanches extend to larger sizes as L grows, with different shapes for topplings, area, loss and reach. Baseline CCDFs use $L = 8, 16, 32, 64$, five seeds and one million inputs per size, conditional on $S > 0$. Left-continuous steps show inclusive $P(X \geq x)$. Zero loss remains in the normalization but is absent on log axes; Table 5.1 gives its mass. Broad ranges alone do not establish universal power laws.

The mean $\mathbb{E}[S]$ in Eq. (3.8) averages over every input, including those that cause no toppling.

Quantity: support	Model	Retained n	α	λ	KS
S: [16,4096]	Power	245,477	1.1426 ± 0.0024	–	0.0310
	Cutoff	245,477	1.0090 ± 0.0082	0.000312 ± 0.000016	0.0074
	Exp.	245,477	–	0.002109 ± 0.000011	0.2468
A: [8,4096]	Power	291,321	1.1354 ± 0.0017	–	0.0461
	Cutoff	291,321	0.9591 ± 0.0023	0.000537 ± 0.000007	0.0185
	Exp.	291,321	–	0.002783 ± 0.000018	0.2717
D: [1,12289]	Power	157,789	1.5098 ± 0.0031	–	0.2242
	Cutoff	157,789	0.2627 ± 0.0312	0.1365 ± 0.0037	0.0247
	Exp.	157,789	–	0.1718 ± 0.0009	0.0468
ℓ: [3,127]	Power	345,680	1.2171 ± 0.0039	–	0.0713
	Cutoff	345,680	0.5817 ± 0.0040	0.0293 ± 0.0002	0.0260
	Exp.	345,680	–	0.0529 ± 0.0003	0.0895

Table 5.2: Discrete marginal fits at $L = 64$, using identical support for the three models within each observable. Parameters are means $\pm$ 95% half-widths across five independent seed fits; n is the total retained count, and KS is the mean descriptive CDF discrepancy. Power fixes $\lambda = 0$ and exponential fixes $\alpha = 0$. The S cap selects an intermediate range; other caps are physical bounds. Loss fits condition on $D > 0$; the other fits use their displayed positive intervals.

On $16 \leq S \leq 4096$, the pure-power effective exponent is 1.1426, with interval $[1.1403, 1.1450]$. Adding an exponential cutoff reduces mean KS from 0.0310 to 0.0074; an exponential alone gives 0.2468. The cutoff model gains 0.0100 nats per retained observation in mean marginal log score over power. Its CDF follows the observed curvature more closely on this range. The comparison is descriptive because the scores use the fitting data and the cutoff model has an extra parameter.

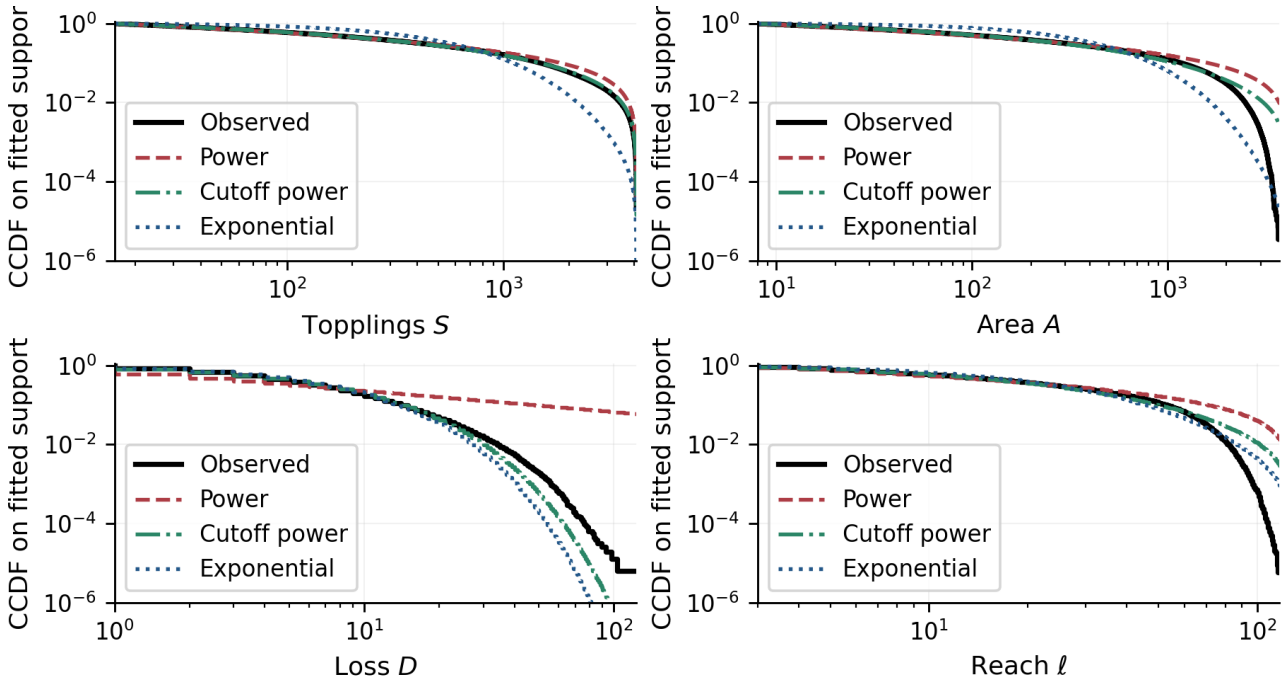


Figure 5.5: Observed and fitted CCDFs at $L = 64$, conditional on the intervals in Table 5.2. Observed curves pool the five equal-length runs. Model curves average the five separately normalized fitted PMFs before cumulation. Normalization uses the full fitting support, even where the display stops near the observed maximum. Cutoff models follow the curves more closely, with some systematic residual shape.

Area also shows curvature: the power and cutoff KS distances are 0.0461 and 0.0185. For positive loss, a pure power gives KS 0.2242, compared with 0.0468 for an exponential and 0.0247 for cutoff power. Reach has a visibly curved finite-support distribution too. The four observables therefore support finite-range descriptions with different shapes.

The estimated slopes depend on the fitting range. At $L = 64$, halving and doubling the chosen lower limits gives pure-power estimates spanning 1.122–1.169 for S, 1.112–1.163 for area, 1.510–1.803 for positive loss and 1.008–1.366 for reach. This variation is separate from the narrower uncertainty intervals conditional on one range. Broad distributions and growing scales match the expected BTW

behaviour, but the range dependence limits interpretation of a single fitted exponent. Earlier work also found that simple scaling fails for BTW toppling statistics [4].

An additional upper-limit check uses the same five $L = 64$ runs and $x_{\min} = 16$, with $x_{\max} = 2048, 4096, 8192$. The retained counts are 231,863, 245,477, 250,246, respectively. Mean cutoff KS distances are 0.0027, 0.0074, 0.0097, compared with 0.0090, 0.0310, 0.0537 for power, and pure-power exponents span 1.092–1.203. Curvature is preferred descriptively at all three caps, while the parameter estimates remain range-dependent.

5.3 Correlations and the role of the source position

We expected a positive correlation between S and A: every site counted in the area must topple at least once, and some sites topple repeatedly. Larger areas should also tend to have greater spatial extent. Loss should depend on these quantities as well as distance from the input site to the boundary. Figure 5.6 supports these predictions. All six Spearman correlations use $S > 0$ and retain zero-loss avalanches.

At $L = 64$, $\rho(S, A) = 0.999188$ with interval $[0.999175, 0.999201]$, $\rho(S, \ell) = 0.9879$ with $[0.9878, 0.9881]$, and $\rho(A, \ell) = 0.9908$ with $[0.9907, 0.9909]$. Repeated topplings and differences in cluster shape allow these observables to differ even when their ranks agree closely. Loss is less strongly correlated: $\rho(S, D) = 0.5349$ with $[0.5321, 0.5377]$, $\rho(A, D) = 0.5332$ with $[0.5304, 0.5360]$, and $\rho(D, \ell) = 0.5237$ with $[0.5206, 0.5267]$.

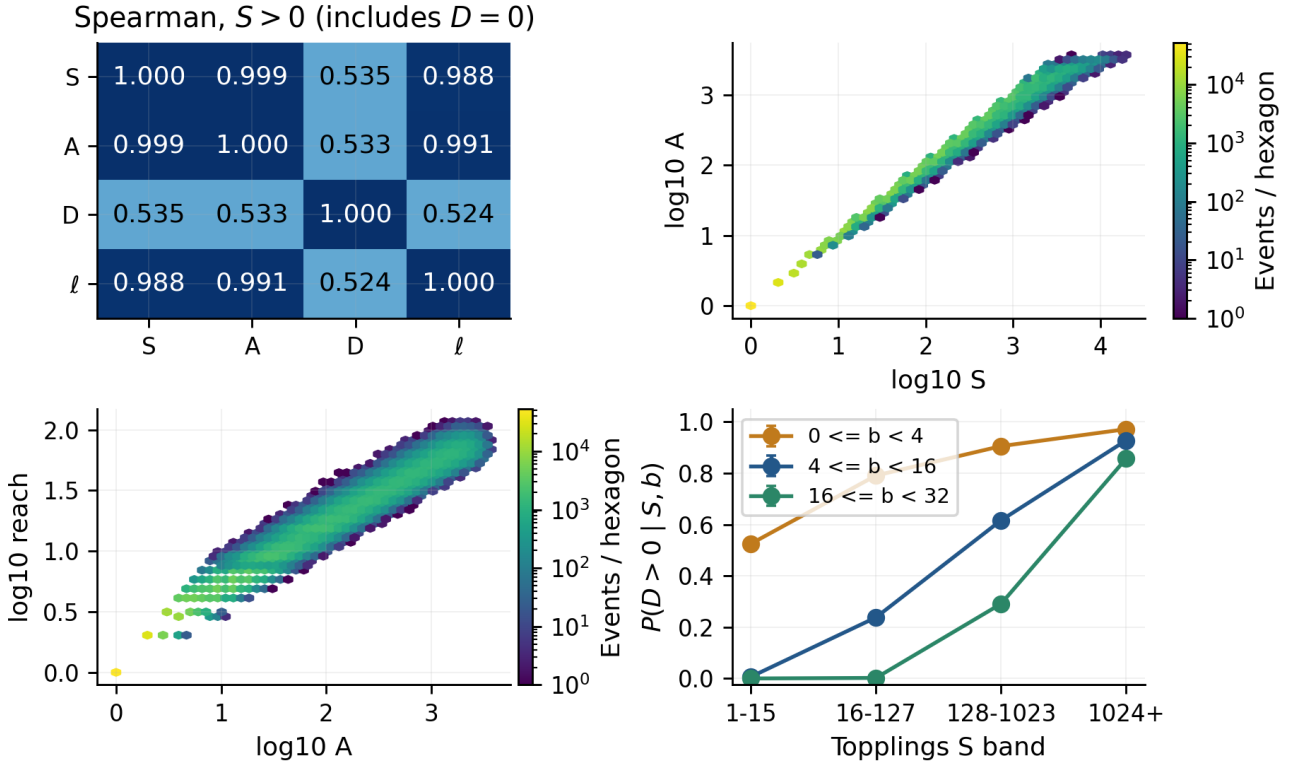


Figure 5.6: Size, area and reach track one another closely; loss depends more on sink access. Baseline $L = 64$ uses 435,058 nonzero avalanches across five seeds, retaining $D = 0$. Top left: mean seed Spearman matrix. Top right and bottom left: joint log-coordinate counts for the full nonzero sample; colour gives events per hexagon. Bottom right: loss probability within fixed size bands and source boundary-distance b strata, with 95% seed intervals.

The boundary-distance groups show that avalanches of similar size are more likely to dissipate when they start nearer a sink. Activity from a source farther inside the grid must reach a boundary before any grain can leave. This helps explain why loss correlates less strongly with topples than area does. Other avalanche features still vary within each group. Each grid size is analysed separately.

5.4 Controlled changes to drive, boundaries and grouping

Table 5.3 compares each extension with its control at the same size. Open and cylindrical runs receive the same grain budget. Batch runs receive the same total grains in a quarter as many events. In each

model, conservation gives stationary mean loss one per grain, while the measurements show how that loss is distributed across events.

L	Condition	Mean height	S/grain	$P(S>0)$	$P(D>0)$	$E[D \mid D>0]$
32	Open/uniform	2.08355 ± 0.00044	40.61 ± 0.07	0.4257	0.2024	4.941
32	Open/biased	2.08297 ± 0.00050	69.02 ± 0.06	0.4409	0.1396	7.166
32	Cylinder	2.10369 ± 0.00059	93.55 ± 0.12	0.4351	0.1687	5.927
32	Batch, $m=4$	2.08355 ± 0.00044	40.61 ± 0.07	0.8917	0.5983	6.686
64	Open/uniform	2.10361 ± 0.00025	153.03 ± 0.43	0.4351	0.1578	6.337
64	Open/biased	2.10313 ± 0.00048	266.67 ± 0.19	0.4442	0.1035	9.662
64	Cylinder	2.11401 ± 0.00053	357.51 ± 0.09	0.4406	0.1271	7.869
64	Batch, $m=4$	2.10361 ± 0.00025	153.03 ± 0.43	0.8978	0.4984	8.025

Table 5.3: Comparisons at matched sizes, with five seeds and one million measured grains per row. Mean height and topplings per grain include 95% seed half-widths. Probabilities and conditional loss use single-grain events for the first three rows at each L and four-grain events for batches. Batch mean loss is near four per event and one per grain. Batch estimates are paired with baseline.

5.4.1 Full-support spatial bias

On the open square, we bias the input towards the centre using

$$q_\eta(x) = \frac{1-\eta}{L^2} + \eta \frac{\exp[-\|x-c\|_2^2/(2\sigma^2)]}{\sum_y \exp[-\|y-c\|_2^2/(2\sigma^2)]}, \quad \eta = 0.9, \quad \sigma = L/8, \quad c = ((L-1)/2, (L-1)/2). \tag{5.1}$$

Every site has positive input probability, bounded below by $0.1/L^2$. Changing these probabilities on the same graph preserves the recurrent stationary measure and its height statistics. The mean response is $\mathbb{E}[S] = q^\top w$, however, so it weights sites according to their input probabilities. Central sites are farther from sinks and have a larger expected response. Biased input can therefore increase mean topplings while preserving the stationary state measure.

At $L = 64$, mean topplings rise from 153.033 to 266.669, close to the actual-q benchmark 266.627 (Figure 5.7). Dissipation probability falls from 0.1578 to 0.1035, while mean loss conditional on dissipation rises from 6.337 to 9.662. Mean loss per grain is 0.9999. The triggering probability can also change: biased input samples sites with different spatial weights even when the stationary law of the grid stays the same.

The initial $L = 32$ height difference was -0.000578, with nominal interval $[-0.001132, -0.000024]$ narrowly excluding zero. At $L = 64$ it was -0.000474, interval $[-0.000948, 0.000001]$. To investigate, we ran a separate confirmation with five new seeds for both models at both sizes, $100L^2$ burn-in and 800,000 measured grains per chain. Each condition received the same larger budget. The differences were 0.000196, interval $[-0.000265, 0.000658]$, and -0.000005, interval $[-0.000234, 0.000224]$, respectively. The longer runs do not reproduce a persistent deficit. With five seeds and several inspected contrasts, the initial nominal interval warrants caution. The confirmation intervals are compatible with zero at the precision of this comparison.

5.4.2 Cylindrical boundaries

For the cylinder, columns are periodic and grains leave only above the top row or below the bottom row. At $L \geq 3$, a toppling on a dissipative row loses one grain, including at the periodic seam. The simulations use widths of at least three. The graph distance is

$$d_{\mathrm{cyl}}((i,j),(i_0,j_0)) = |i-i_0| + \min(|j-j_0|, L-|j-j_0|), \tag{5.2}$$

and the source distance to a dissipative boundary is $\min(i, L-1-i)$. Reach follows this cylindrical distance and is bounded by $L + \lfloor L/2 \rfloor$.

A finite-grid benchmark follows by taking w constant along each row. Set $w_{-1} = w_L = 0$ and

$$w(i,j) = \frac{(i+1)(L-i)}{2}. \tag{5.3}$$

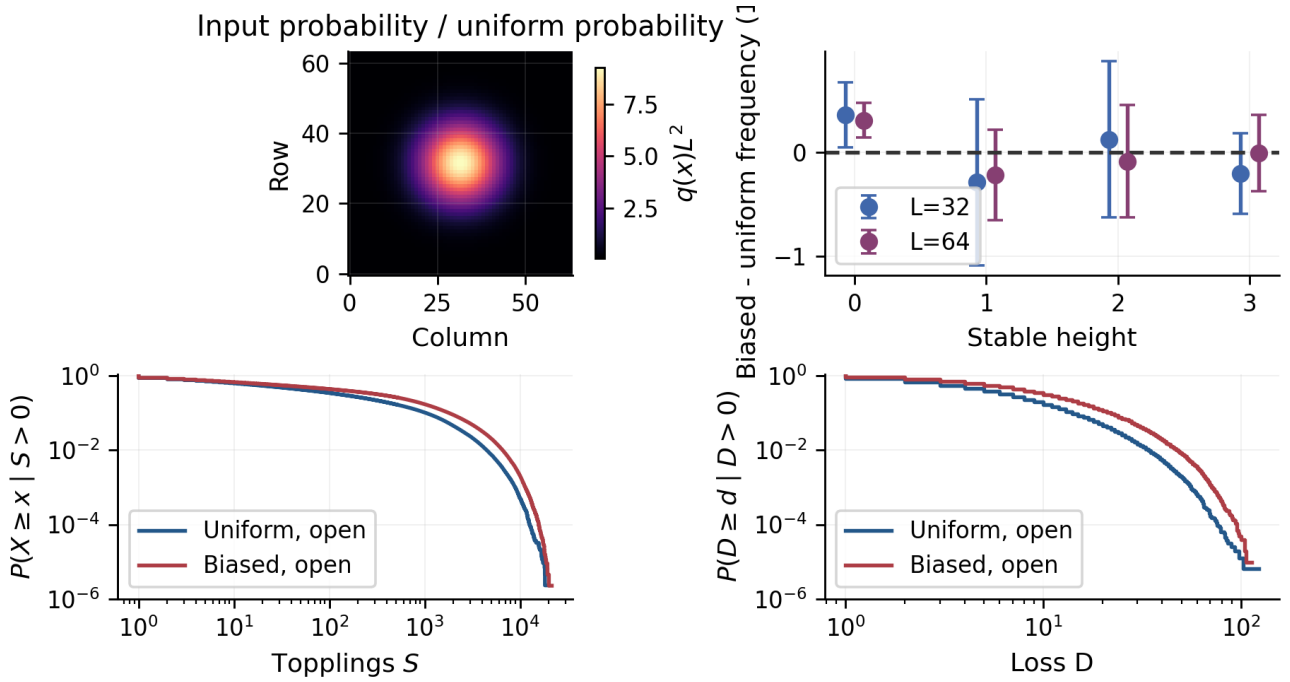


Figure 5.7: Spatial bias changes avalanche responses; the stationary height distribution is theoretically invariant on the same graph under full-support driving. Top left: $L = 64$ input probability relative to uniform. Top right: finite-sample biased-minus-uniform height-frequency differences at $L = 32, 64$, with Welch 95% independent-seed intervals; the theoretical difference is zero. Bottom: $L = 64$ size ($S > 0$) and loss ($D > 0$) CCDFs, with five seeds per condition.

The horizontal neighbours cancel two diagonal contributions, giving $(K_{\text{cyl}}w)_{i,j} = 2w_i - w_{i-1} - w_{i+1} = 1$, including on the dissipative rows. Averaging over uniform input yields

$$\mathbb{E}[S]_{\text{cyl}} = \frac{1}{L}\sum_{i=0}^{L-1}\frac{(i+1)(L-i)}{2} = \frac{(L+1)(L+2)}{12}. \tag{5.4}$$

Substitution into the sparse matrix verifies this analytic result.

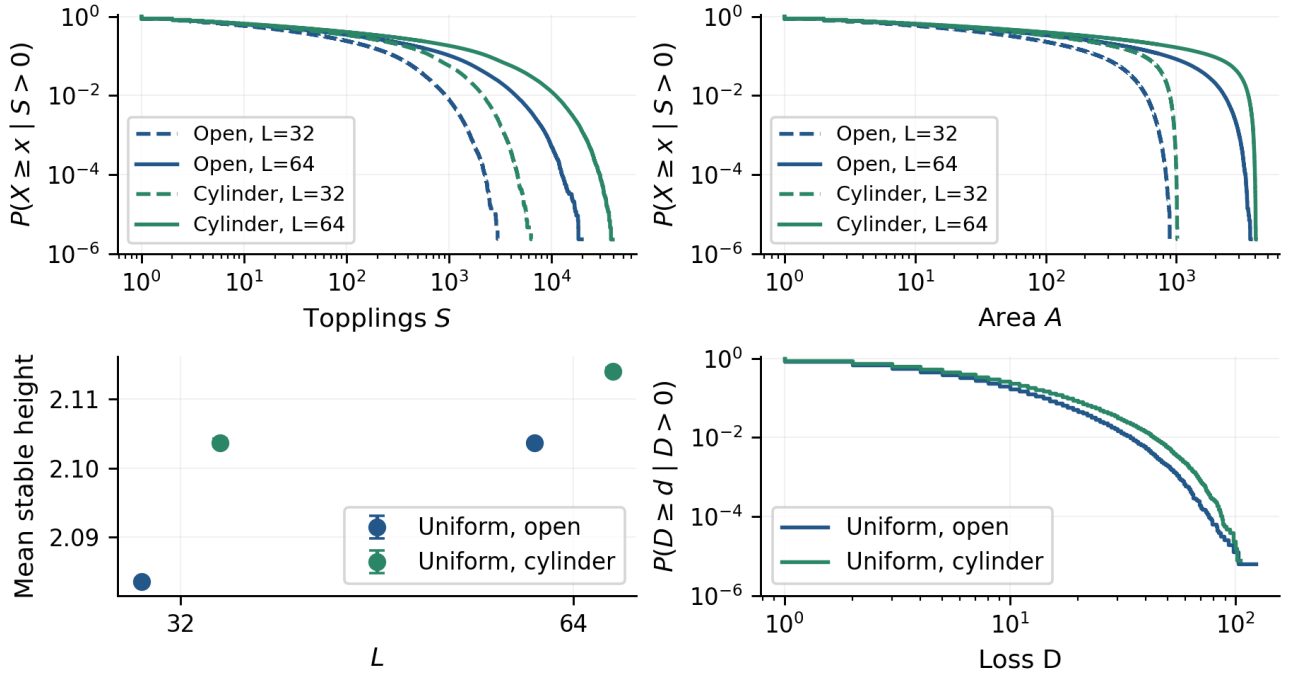


Figure 5.8: Cylinder and open boundaries. Top: size and area CCDFs at $L = 32, 64$, conditional on $S > 0$. Bottom left: mean stable height with 95% seed intervals. Bottom right: positive-loss CCDFs at $L = 64$. Each curve uses five primary seeds. Both models receive one grain per event and have theoretical mean loss one, with different loss frequencies and conditional magnitudes.

Equation (5.4) predicts 93.5 and 357.5 topplings at $L = 32, 64$. The measured means are 93.547

and 357.511; at $L = 64$, the interval $[357.416, 357.605]$ contains the benchmark. Mean height exceeds the open control by 0.010406, interval $[0.009887, 0.010925]$. Here the boundary change affects the height statistics as well as the response. With fewer dissipative edges, grains stay longer and generate more topplings, while conservation still fixes stationary loss at one per input. At $L = 64$, mean loss is 1.0000, with dissipation probability 0.1271 and conditional mean loss 7.869. Figure 5.8 shows how the boundary change affects size, area and loss.

5.4.3 Batches as an exact grouping experiment

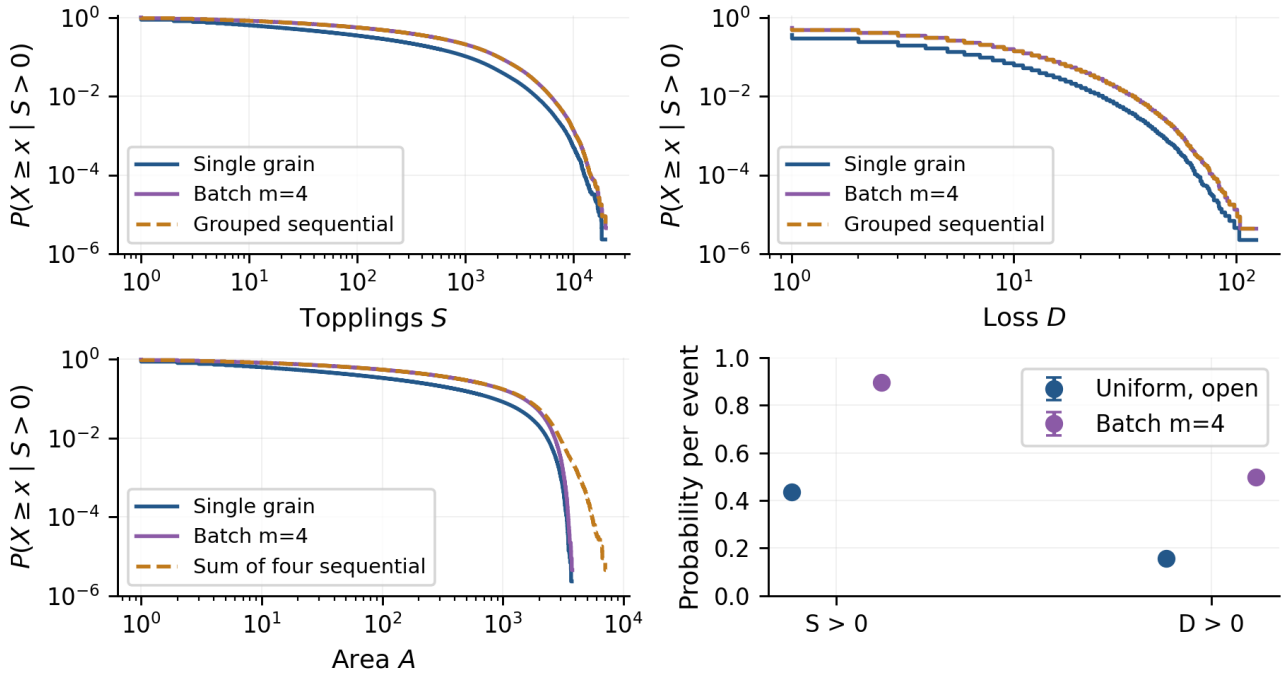


Figure 5.9: Event grouping at $L = 64$ for five matched runs: one million baseline single-grain events and 250,000 four-grain batches. Grouped sequential size and loss coincide with the batch values. For area, the third curve sums four sequential areas, counting shared sites repeatedly; the batch counts their union. Curves condition on nonzero grouped or ungrouped avalanches as appropriate. Bottom right: triggering and dissipation probabilities with seed intervals. Per-grain mean topplings and loss agree exactly in the paired runs.

Each batch draws four independent uniform sites with replacement and adds their grains before relaxation. We compare this with sequential single-grain additions from the same stable h, using the same ordered sites $s_1, \ldots, s_m$. A toppling in the sequential schedule remains legal if later, nonnegative inputs are already present. Concatenating those schedules therefore stabilizes the batched input legally. Uniqueness of the odometer gives

$$h_{\text{batch}} = h_{\text{sequential}}, \quad u_{\text{batch}} = \sum_{r=1}^{m} u_r, \quad S_{\text{batch}} = \sum_{r} S_r, \quad D_{\text{batch}} = \sum_{r} D_r. \tag{5.5}$$

Area counts the union of toppled sites. The sum $\sum_r A_r$ counts shared sites more than once, whereas the toppling and loss identities hold even for correlated consecutive avalanches.

If P is the one-grain transition, independent sources give batch transition P^m. Every stationary measure of P is stationary for P^m. Depending on the sampling period, P^m can still have more than one recurrent class or fail to converge from a particular start. We use a matched baseline grid after equilibration so that the comparison isolates grouping. Each batch is fully relaxed before the next one arrives.

Across both sizes, all 500,000 batches match the grouped controls exactly in S, D and final mass. Grid configurations and height frequencies also agree whenever sampled. Independent small-grid tests verify the full odometer and area-union identities. At $L = 64$, 12.01% of batches have overlapping toppled sets, and summing sequential areas exceeds the mean union by 16.64 sites per batch. These overlaps account for the difference between the two area curves.

Grouping raises toppling probability from 0.4351 to 0.8978 and loss probability from 0.1578 to 0.4984. Mean loss is about four per batch, with exactly the baseline value per grain. Combining successive, potentially overlapping single-grain responses changes event distributions.

Chapter 6

Discussion, limitations, and conclusion

A simple deterministic local toppling rule produces broad avalanche responses as inputs encounter different configurations. The baseline has unequal stable-height frequencies and finite-grid mean topplings that agree with analytic predictions. Size, area and reach have closely related ranks, while loss depends more on access to a sink.

The three extensions distinguish changes in state, response and event definition. On a fixed graph, full-support driving preserves the stationary recurrent measure but changes how input sites weight the response. Cylindrical boundaries alter connectivity and dissipation, changing the measured stationary heights and mean topplings. Their response is governed by the inverse toppling matrix K^{-1}, which maps mean input to mean sitewise topplings in Eq. (3.8). Batching changes event distributions while preserving grouped sequential toppling totals, loss totals and final configurations exactly. Its larger events combine successive, potentially overlapping responses.

The evidence is limited to four baseline sizes, two sizes per extension and five primary seeds per condition. Temporal dependence motivates seed-level and block-based uncertainty estimates, whose precision is limited by the replicate count. Stationarity diagnostics concern selected observables; the confirmation examines one height discrepancy. Finite-range marginal fits have systematic shape errors and range-dependent parameters, so their exponents describe these samples. KS and score comparisons are descriptive, without calibrated IID goodness-of-fit p-values. Batches relax completely between inputs, which leaves finite-rate driving untested. Performance gains were measured on one workload; total simulation times include the complete runs.

Conservation fixes long-run mean loss at one per input grain, despite changes in loss frequencies, magnitudes and event grouping. Growing finite-size ranges and better cutoff descriptions support broad, size-limited responses; range-dependent exponents and residual shape do not establish universal power laws. A changed avalanche distribution must therefore be interpreted through the stationary configuration measure, the response to driving and the definition of an event.

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